De La Salle University - Dasmariñas
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-ITCS318LC
COURSE TITLE	**:**	INFORMATION ASSURANCE AND SECURITY
COURSE TYPE	**:**	Laboratory
COURSE CREDIT	**:**	1 Unit
MODALITY	**:**	**Fully Onsite**
PRE-REQUISITES	**:**	S-ITCS123, S-ITCS123LA
CO-REQUISITES	**:**	S-ITPC318
PROFESSOR	**:**	MARIVIC R. MITSCHEK
CONSULTATION HOURS:		M(700AM-1000AM-PCH101), T (700AM-1200NN-ONLINE) mrmitschek@dlsud.edu.ph

VISION

A leading Catholic university that inspires excellence and drives innovation towards a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension program.

COURSE DESCRIPTION:

This course examines fundamentals of network security involved in creating and managing secure computer network environments. Both hardware and software topics are considered including authentication methods, remote access, network security architectures and devices, cryptography, forensics, and disaster recovery plans. This course adopts the two courses included in the online curriculum of Junior Cybersecurity Analyst Path of the Cisco Networking Academy including Introduction to Cybersecurity and Endpoint Security.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. Understand the players in the cybersecurity world and the motivation of cyber criminals and cybersecurity specialists;
CLO2. analyze security attacks, symptoms, processes, and countermeasures;
CLO3. apply skills in security management, controls, protection, and mitigation technologies;
CLO4. illustrate how cybersecurity professionals use technologies, processes and procedures to defend all components of the network; and
CLO5. create a security measure plan for the different Cybersecurity domains.
CLO6. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Understand the players in the cybersecurity world and the motivation of cyber criminals and cybersecurity specialists.	TLO1. Describe the common characteristics comprising the cybersecurity world TLO2. Understand the three principles of information assurance and security.
CLO2. Analyze security attacks, symptoms, processes, and countermeasures.	TLO3. Differentiate the types of malwares and malicious code. TLO4. Compare the different methods used in social engineering.
CLO3. Apply skills in security management, controls, protection, and mitigation technologies.	TLO5. Explain how encryption techniques protect confidentiality. TLO6. Explain processes used to ensure integrity. TLO7. Explain the concept of high availability.
CLO6. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	**Week**	**Teaching Learning Activities/Assessment Strategies**	**Technology Enabler**	**Allotted Hours**

Introduction Module	1 Aug 19 - Aug 24	Syllabus Uploaded Gospel reading, prayer and reflection Presentation of course introduction Course requirement details	Schoolbook	3
Module 1 TLO 1 CLO 1 CLO 6	2 Aug 26 - Aug 31	Chapter 1: Introduction to Cybersecurity Gospel reading, prayer and reflection Enabling Assessment: What was Taken? Cybersecurity Job Hunt Chapter 2: Attacks, Concepts and Techniques Packet Tracer 3.3.2.7 - WEP/WPA2 PSK/WPA2 RADIUS	Schoolbook, Packet Tracer	3
Module 2 TLO 2 CLO 1 CLO 6	3 Sept 2 - Sept 7	Chapter 3: Protecting your Data and Privacy Gospel reading, prayer and reflection Enabling Assessment: Who owns your data? Packet Tracer 2.5.2.7 - Using File and Data Integrity Checks Discover your own Risky online Behavior Chapter 4: Protecting the Organization • Lab Activity 4.3.2.3 – Using Steganography Chapter 5: Will Your Future be in Cybersecurity	Schoolbook, Packet Tracer	3
Module 1, 2, 3	4 Sept 9 - Sept 14	Summative Exam on Introduction to Cybersecurity Course • Packet Tracer 4.3.3.3– Configuring VPN Transport Packet Tracer 4.3.3.4– Configuring VPN Tunnel Mode		3
Module 4	5 Sept 16 - Sept 21	Governance and Compliance Gospel reading, prayer and reflection Enabling Assessment Developing Cybersecurity Policies and Procedures Network Security Testing Enabling Assessment Packet Tracer – Use Diagnostic Commands Use Wireshark to Compare Telnet and SSH Traffic		3
Module 5	6 Sept 23 - Sept 28	Threat Intelligence Gospel reading, prayer and reflection Enabling Assessment Lab - Evaluate Cybersecurity Reports Lab - Identify Relevant Threat Intelligence Endpoint Vulnerability Assessment		3

		Lab – Risk Management Lab – Risk Analysis Lab – Security Controls Implementation		
Module 6	7 - Sept 30 - Oct 12	Digital Forensics & Incident Analysis & Response Gospel reading, prayer and reflection Lab – Gather System Information After and Incident Lab – Attack Analysis Lab – Incident Handling Packet Tracer – Investigate Disaster Recovery		3
Module 4, 5, 6	8 Oct 14 - Oct 19	Cyber Threat Management Exam		***3***
			TOTAL	***24.0***

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO3. Apply skills in security management, controls, protection, and mitigation technologies.	TLO6. Explain processes used to ensure integrity. TLO7. Explain the concept of high availability.
CLO4. Explain how cybersecurity professionals use technologies, processes, and procedures to defend all components of the network.	TLO8. Explain how processes and procedures protect systems. TLO9. Explain how to implement security measures to protect network devices.
CLO5. Create a security measure plan for the different Cybersecurity domains.	TLO10. Document the security measure plan for the different Cybersecurity domains.
CLO6. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	**Week**	**Teaching Learning Activities/Assessment Strategies**	**Technology Enabler**	**Allotted Hours**

Module 7	**10** Oct 21 - Oct 26	• Cybersecurity Threats, Vulnerabilities and Attacks Part 1 Gospel Reading, Prayer and Reflection	Schoolbook, Packet Tracer	**3**
	11 Oct 28 - Oct 30	**Independent Learning Days**		
Module 7 TLO 6 TLO 7 CLO 3 CLO 6	12 Nov 4 - Nov 9	• Cybersecurity Threats, Vulnerabilities and Attacks Part 2 • Gospel Reading, Prayer and Reflection	Schoolbook, Packet Tracer	3
Module 8 TLO 10 CLO 5 CLO 6	13 Nov 11 - Nov 16	Attacking the Foundation Gospel Reading, Prayer and Reflection • Lab – Reading Server Logs • Lab – Recommend Threat Mitigation Measures	Schoolbook, Packet Tracer	3
Module 9 TLO 6 TLO 7 CLO 3 CLO 6	14 Nov 18- Nov 23	Wireless Network Communication Enabling Assessment Gospel Reading, Prayer and Reflection o Network Security Infrastructure o Packet Tracer – ACL Demonstration	Schoolbook, Packet Tracer	3
Module 10 TLO 6 TLO 7 CLO 3 CLO 6	15 Nov 25-30	**Checkpoint Exam**	Schoolbook, Packet Tracer	3
Module 11 TLO 8 TLO 9 CLO 4 CLO 6	16 Dec 2-7	The Windows Operating System Enabling Assessment Gospel Reading, Prayer and Reflection o Lab – Monitor and Manage System Resources in Windows Cybersecurity Principles, Practices and Processes •	Schoolbook, Packet Tracer	3

Module 12 TLO 8 TLO 9 CLO 4 CLO 6	17 Dec 9-14	• Linux Overview • System and Endpoint Protection Enabling Assessment o Lab – Harden a Linux System o Lab – Recover Passwords o Lab – Online Malware Investigation Tools	Schoolbook, Packet Tracer	3
Total				24

GRADING SYSTEM

Midterm

Enabling Assessments	- 40%
Summative Assessments	- 60%
	100%

Finals

Enabling Assessments	- 40%
Summative Assessments	- 60%
	100%

Overall Final Grade = $\frac{\text{Midterm + Final}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Onsite Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:

i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
iii. Email – you may email your professor. (Professor's email will be given during the course orientation)

c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
   b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES

Online References

Call number or e-provider	**Reference Material**
www.netacad.com	Cisco Networking Academy, [accessed 2024 August 15] Retrieved from https://www.netacad.com/portal/resources/course-resources/cybersecurity-essentials
2399513	Gupta CP, Goyal KK. 2020. *Cybersecurity : a self-teaching introduction.* Dulles, Virginia: Mercury Learning And Information.

1849580	Hulick K. 2019. Cybersecurity careers. San Diego, CA: ReferencePoint Press, Inc.
10.1145/3527202	Wroclawski J, Benzel T. 2022. Cybersecurity as illuminator for the future of computing research. Communications of the ACM. 65(5):39–41. doi:10.1145/3527202.

Onsite References

Call number or eprovider	**Reference Material**
KF1263.C65 .K847 2017	Kosseff J. 2019. *Cybersecurity law*. Hoboken: Wiley.

Prepared: **Information Technology Department**
1st Semester 2024-2025

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